



# A cooperative EV charging scheduling strategy based on double deep Q-network and Prioritized experience replay

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## ABSTRACT

The rapid growth of the number of electric vehicles (EVs) has significantly increased the demand for electricity of residents, which may lead to transformer overload of distribution networks if there is no reasonable power consumption planning. In view of the coordinated charging of EVs under the photovoltaic (PV) energy connected to the power grid, a collaborative charging control strategy is proposed based on a double deep q-network with Prioritized experience replay (DDQN-PER). Specifically, Long-Short-Term Memory (LSTM) neural network is utilized to capture the uncertainties caused by power requirement, PV power, and real-time electricity price. Then, a modified DDQN approach and the corresponding reward function are applied to solve the cooperative charging problem. In addition, the introduction of a PER mechanism based on TD bias sampling in reinforcement learning alleviates the problem of reward sparsity in learning scenarios and improves the training stability and efficiency. The proposed approach can achieve collaborative scheduling of EVs, which helps reduce the charging cost, promote the consumption of PV energy and avoid transformer overload. Simulation case studies based on real data demonstrate that the proposed algorithm can reduce the charging cost by 30% and increase the PV power utilization by 10% compared to the DDQN algorithm.

## 1. Introduction

### 1.1. Motivation

In recent years, with the advances in technology and the enhancement of public awareness of environmental protection, the market share of electric vehicles (EVs) is getting higher and higher. The wide adoption of EVs can significantly reduce the consumption of fossil fuels in the field of transportation, and realize the green and sustainable development of transportation (Chen et al., 2021). Although the large-scale integration of EVs has brought many benefits, it will undoubtedly have a great impact on the safe operation of the distribution network. Uncontrolled charging of large-scale EVs will lead to distribution transformer overload and power quality decline, which is not conducive to the safety, stability, and economic operation of the power grid, and may also cause road congestion around charging stations and reduce the traffic efficiency (Galus et al., 2012). Hence, it is significant to optimize and control the charging process of EVs. EV charging load is a typical schedulable load (Maigha and Crow, 2017), and at the same time, EVs are also a kind of distributed power source. Optimizing EVs' charging/discharging process can not only achieve peak load shifting and smooth load curve (Silva et al., 2020a; Nunna et al., 2018), promote the consumption of renewable energy (Huang and Wang, 2021a);

but also provide auxiliary services such as frequency modulation and spinning reserve for the grid; and it can also reduce the charging cost of EV users (Yao et al., 2017a). However, due to the uncertainties in traffic conditions, user's commuting behavior, renewable energy's power output, transformer base load, and the variation of real-time electricity price, it is challenging to optimize the charging/discharging process of EVs while guaranteeing distribution transformer will not be overloaded and maximizing the utilization of renewable energy.

### 1.2. Literature review

The essence of optimal scheduling of EVs is to arrange the timing of charging/discharging state considering various uncertain factors. Most algorithms proposed in this field are model-based (Pillai and Bak-Jensen, 2011; Huang et al., 2015). Traditional EV charging strategies based on optimization/robust optimization require iterative computation and substantial computational resources. The performance is largely dependent on the prior knowledge of the system. Compared with these traditional optimization/robust optimization strategy, reinforcement learning (RL) can learn the optimal control strategy through the interaction between agent and environment, which need not the accurate system model and state transfer function. RL can efficiently solve

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sequential decision problems in complex and uncertain environment, such as EV charging/discharging optimization.

Many studies are devoted to using model-free deep RL to coordinate the EV charging/discharging processes to reduce the user charging costs (Zhang et al., 2020) and avoid transformer overload (Wang et al., 2020). In Wan et al. (2019), a model-free method based on deep-Q-network (DQN) is proposed to determine the optimal charging strategy, using a Q network approaching the optimal action value function to adaptively learn the transfer probability without any system model information. The above methods based on conventional DQN relies heavily on the design of penalty terms and the selection of penalty coefficients. In order to avoid the error caused by the manual selection of reward and punishment coefficient, in Li et al. (2020), an improved DQN named safety deep reinforcement learning (SDRL) is proposed, which directly learns the original state information and electricity price information of EVs through the deep neural network, to generate a constrained optimal charging/discharging schedule. To consider the scheduling optimization under the continuous action space, in Zhang et al. (2021), an improved control deep deterministic policy gradient(CDDPG) algorithm is proposed to learn the optimal charging control strategy of EVs, so as to minimize the charging cost of EV users on the premise of meeting their energy demand.

In the above studies, the fluctuation of transformer base load on the optimization results are not considered. In actuality, the total EV charging load and transformer base load may exceed the maximum capacity of the distribution transformer, at evening when the base load is high. To eliminate the negative impact of uncoordinated EV charging on distribution network and minimize user charging cost. Cao et al. (2022) proposes an intelligent charging algorithm based on actor-critical (AC) learning. Silva et al. (2020b) proposes a multi-agent, multi-object reinforcement learning structure (MASCO) for EV charging, where MASCO makes minimal assumptions about the distribution network, works at any type of electricity price, and can be configured according to consumer preferences. Yan et al. (2022) proposes a collaborative charging control strategy based on multi-agent deep reinforcement learning (MADRL) for EVs. The proposed method is capable of solving continuous states and actions. The local observations and approximately training agents show that the proposed method is fully decentralized and scalable for the multi-agent problem. Li et al. (2022) proposes a distributed transformer joint optimization method based on MADRL, the scheduling of multiple EVs is formulated as a Markov decision problem (MDP), the method can coordinate EV charging based on local information, and it is propitious to protect the privacy of EV Users, reduce communication equipment deployment cost, avoid a single point of risk. Unlike the above papers, which did not consider renewable energy, Huang and Wang (2021b) proposes a deep reinforcement learning (DRL) agent based on proximal strategy optimization (PPO) to perform capacity scheduling for PV cell energy storage systems. With training, PPO agents are able to adapt to highly uncertain and volatile market signals and PV power generation profiles. The authors of Shin et al. (2020) proposed a distributed MADRL based energy management algorithm for EV charging station considering the influence of PV system and energy storage system. But, the coordination of renewable energy system and transformer load and the utilization of renewable energy are not considered.

Integrating renewable energy to the grid can effectively alleviate the peak load of transformers, reduce charging costs, and promote its utilization. However, the output of renewable energy is intermittent, and it is a big challenge to coordinate the charging process of a large number EVs to reduce charging cost, maximize renewable energy utilization rate and avoid distribution transformer overload simultaneously in such a uncertain environment.

### 1.3. Contribution of this paper

Inspired by these researches, considering the randomness of commuting and PV generation, the fluctuation of electricity price and the capacity limit of distribution transformer, this paper proposes a EV collaborative charging control strategy, in order to reduce the charging cost of EV users, promote renewable energy utilization, and eliminate the negative impact of EV charging on distribution transformer. The EV charging scheduling problem is formulated as a MDP problem, and it is solved by a state of the art DRL algorithm named Double Deep Q-Network and Prioritized Experience Replay (DDQN-PER). Firstly, three feature extraction networks based on Long Short Time Memory (LSTM) are used to extract the time characteristics of electricity price, basic load and PV power generation. After that, the extracted features are fed into a Q network to learn the optimal action value function. After training, the optimal EV charging/discharging strategy can be derived and used to coordinate EV charging process. To strike a good balance between computational complexity and algorithm performance, the charging power is discretized. DDQN-PER is a modified DQN algorithm that introduces a Prioritized Experience Replay (PER) (Schaul et al., 2015) based on DDQN (Hasselt et al., 2016), which prioritizes the sample empirical selection by Temporal Difference(TD) bias values. The algorithm is tested using real data and compared with other baseline algorithms. Experimental results show that the proposed algorithm can effectively reduce the charging cost, promote the use of renewable energy, and avoid the distribution transformer overload. Furthermore, to make the optimization results as realistic as possible, the cost of battery degradation is also considered. The contributions of this article can be summarized as follows:

- Considering the randomness of users' commuting and PV power generation, the fluctuation of electricity price and the capacity limit of distribution transformers, a collaborative charging/discharging network of multiple EVs and PV systems with unknown transmission probability is established.
- Three feature extraction network based on LSTM are constructed to extract the characteristics of electricity price, basic load and PV power generation. These features combined with the battery SoC, EV's available charging time, and electricity price are fed to the Q network to learn the optimal action value function.
- A real-time EV charging/discharging collaborative control algorithm based on DDQN-PER is proposed, and the priority is determined by selecting the sample experience through the TD error value. Setting the appropriate reward function to improve the training stability and efficiency.

### 1.4. Organization of the paper

The rest of this paper is organized as follows: Section 2 describes the problem and formulates it as an MDP model with unknown transition probability. The proposed real-time EV charging/discharging algorithm based on DDQN-PER is described in detail in Section 3. Section 4 discusses the experimental setup and simulation results. Section 5 concludes the paper.

## 2. Problem formulation

Charging/discharging scheduling of EV is to control the charging/discharging actions of charging pile to EV at each time step. The scheduling horizon is divided into  $N$  time steps, and each time step is  $\Delta t$  in hour. During the scheduling process, the charging/discharging action is updated every time step, which is a typical sequential decision problem. This problem can be formulated as a finite MDP model  $(S, A, P, R, \gamma)$  with discrete time step, where,  $S$  represents the set of all states in the environment;  $A$  represents the set of all actions that can be taken by the agent, and each action  $a_t$  represents the charging/discharging power of the EV;  $P$  indicates the probability of

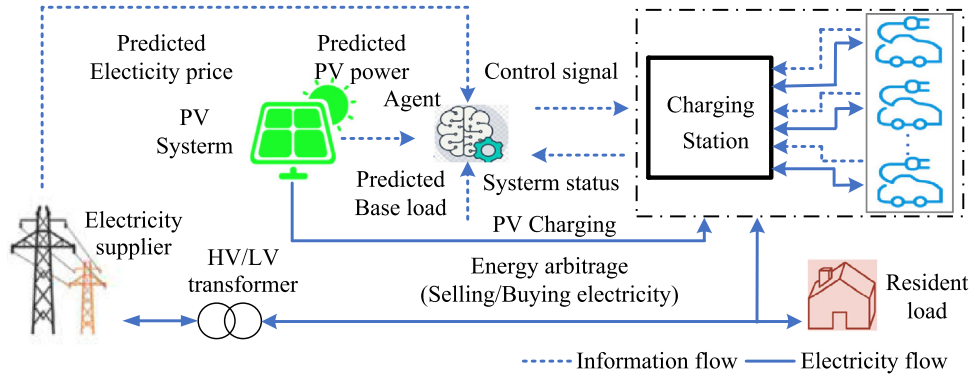


Fig. 1. Schematic diagram of the cooperative charging/discharging scheduling framework.

state transition;  $R$  is the reward function, i.e.,  $R(s, a)$  is the immediate reward for agent when taking action  $a$  in state  $s$ ;  $\gamma \in [0, 1]$  is a discount factor. At time step  $t$ , the agent observes the environment state  $s_t$ , then selects a charging/discharging action  $a_t$ . After executing this action, the environment state transmits to a new state  $s_{t+1}$  and the agent gets a reward  $r_t$ . The algorithm proposed in this paper aims to find an optimal policy to maximize the long-term cumulative reward of EV users.

Fig. 1 is the schematic diagram of the cooperative charging/discharging scheduling framework proposed in this paper, where, both the PV system and main grid participate in EV charging behaviors. The intelligent dispatching agent schedules the battery capacity to meet the power requirements of users, minimize the cumulative charging cost and avoid transformer overload by processing market signals, such as PV power, EV status, basic load, and real-time electricity price.

As above mentioned, the EV charging/discharging processes are formulated using MDP model. Next, the details of the MDP development are shown below.

(1) **State**: The environment at time step  $t$  is represented by a vector of 6 variables:

$$s_t = (p_t, \lambda_t, e_t, l_t, P_{\text{base},t}, P_{\text{pv},t}) \quad (1)$$

where,  $p_t$  represents the electricity price at the current moment  $t$ ;  $\lambda_t$  is PV power price;  $e_t$  represents the SOC of the EV battery;  $l_t$  is the remaining charging/discharging time slots:  $l_t = t_{\text{end}} - t$ , it is the departure moment chosen by the user minus the current moment. When the remaining charging time slot is 0, it means that the electric vehicle leaves the charging station after completing the charging task.  $P_{\text{base},t}$  denotes the base load;  $P_{\text{pv},t}$  is the PV power.

(2) **Action**: Action  $a_t$  is the EV charging/discharging power at time step  $t$ .  $a_t$  is positive when EV is in charge action and negative when it is in discharge action.  $a_t = 0$  indicates that EV is in idle state with neither charge nor discharge. Therefore, EV users can fully consider the situation of real-time electricity price and take charge action when the electricity price is low and discharge action when the electricity price is high, so as to earn a certain profit and minimize their charging cost. Ideally, the charging power should be continuously adjustable, but it would increase the computational burden. Therefore, the charging power is discretized in this study as that in Le Floch et al. (2018), Sun et al. (2018). The selection of the levels of discretization  $n$  is a compromise between the flexibility of charging power and the computational complexity. Without loss of generality, to simplify the computation, the charging/discharging power of EVs is discretized as follows:

$$a_t \in \left\{ -P_{\text{max}}, -\frac{n-1}{n}P_{\text{max}}, \dots, 0, \dots, \frac{n-1}{n}P_{\text{max}}, P_{\text{max}} \right\} \quad (2)$$

where  $P_{\text{max}}$  is the maximum charging power,  $-P_{\text{max}}$  is the maximum discharging power, and  $n$  is a parameter that represents the total grade number of charging/discharging power.

(3) **State transition**: The state transition from  $s_t$  to  $s_{t+1}$  is determined by

$$s_{t+1} = f(s_t, a_t, w_t) \quad (3)$$

The state transition is not only controlled by the action  $a_t$  but also influenced by some random factors such as EV arrival time, remaining charging time, transformer base load, PV output energy and electricity price. In Eq. (3), symbol  $w_t$  is used to represent these random factors. Specifically, the battery residual energy  $e_t$  is affected by the charging/discharging power  $a_t$ , which can be accurately expressed by the deterministic model  $e_{t+1} = e_t + a_t \cdot \Delta t$ . It is assumed that there is no power loss in EVs charging, without loss of generality, as shown in Wan et al. (2019), Li et al. (2020). However, other components in state vector are random, and it is difficult to find an accurate distribution model of  $w_t$ . To solve this problem, a DDQN-PER based EV collaborative charging scheduling strategy is proposed in this paper to learn the state transformations, as detailed in the next section.

(4) **Reward**: Assuming that there are  $M$  EVs for charging. Each EV  $i$  will receive a local reward  $r_t$  when the global state changes from  $s_t$  to  $s_{t+1}$  with the action  $a_t$ . The reward function is associated with the control goal. In this paper, the cooperative control objective is to minimize charging cost (including battery degradation cost) and RA while avoiding transformer overload and improving PV utilization. Therefore, the reward function consists of three parts: the penalty of charging cost, the penalty of RA and the penalty for the transformer overload.

$$r_t = \begin{cases} -p_t(a_t^i - p_{\text{pv},t}^i)\Delta t - \lambda_t p_{\text{pv},t}^i \Delta t - c_{\text{bat},t}^i - \beta P_{\text{overload},t}^i & -\tau \delta_t^i (e_{\text{max}} - e_t), P_M^t > P_{\text{pv},t} \\ -\lambda_t p_{\text{pv},t}^i \Delta t - c_{\text{bat},t}^i - \beta P_{\text{overload},t}^i - \tau \delta_t^i (e_{\text{max}} - e_t), 0 \leq P_M^t \leq P_{\text{pv},t} \end{cases} \quad (4)$$

where  $P_M^t = \sum_{i=1}^M a_t^i$  is the total charging load of all EVs at time step  $t$ . As described in Eq. (4), there are two situations according to the power output of PV. Firstly, if the total charging load  $P_M^t$  is greater than the PV power output  $P_{\text{pv},t}$ , all energy generated by PV is for EV charging, and the shortage  $P_M^t - P_{\text{pv},t}$  is supplied by the main grid. Second, if the PV power output is greater than the total EV charging load, all EV charging demand is met by PV.  $p_{\text{pv},t}^i \geq 0$  represents the charging power that EV  $i$  gets from PV at time step  $t$ ; if EV is discharging or idle, i.e. when  $a_t^i \leq 0$ ,  $p_{\text{pv},t}^i = 0$ . The first term  $p_t(a_t^i - p_{\text{pv},t}^i)\Delta t$  and the second term  $\lambda_t p_{\text{pv},t}^i \Delta t$  indicate the charging cost of EV  $i$  at time step  $t$ , where  $\lambda_t$  is the price of PV power. If EV is discharging, the second term is zero, and the first term will turn to  $p_t a_t^i \Delta t < 0$ . When  $0 \leq P_M^t \leq P_{\text{pv},t}$ , the total charging cost is  $-\lambda_t p_{\text{pv},t}^i \Delta t$ . As suggested by Ortega-Vazquez (2014a), Sarker et al. (2015), we assume that the price of electricity sold to the grid is the same as the price of electricity purchased.

When the PV power output is insufficient, how to allocate the limited PV energy to different EVs is very important. The allocated charging power from PV for EV  $i$  at time step  $t$  is decided by Eq. (5):

$$p_{\text{pv},t}^i = \begin{cases} \frac{a_t^i}{\sum_{i=1}^M a_t^i} \cdot P_{\text{pv}}^t & a_t^i > 0 \\ 0 & a_t^i \leq 0 \end{cases} \quad (5)$$

when  $a_t^i > 0$ , the PV power  $p_{\text{pv},t}^i$  that the vehicle can access is determined by the weight of the current charging power in the total power

of all charging vehicles; when  $a_i^t \leq 0$ ,  $p_{pv,i}^t$  is 0.  $c_{bat,i}^t$  represents the cost of battery degradation during charging/discharging of EV, and the degradation cost is calculated by Eq. (6) Ortega-Vazquez (2014b):

$$c_{bat,i}^t = C_{batCost} \left| \frac{m_k}{100} \right| \frac{a_i \times \Delta t}{e_{max}} \quad (6)$$

where  $C_{batCost}$  represents the total cost of battery purchase,  $m_k$  represents the approximate slope of the linear relationship with the number of charging/discharging cycles, numerical values are obtained from Ortega-Vazquez (2014b). And  $e_{max}$  represents the maximum available capacity of the battery. The penalty value of the EV charging cost and the battery degradation cost is calculated according to the charging decision and the EV battery energy level.

Different from user charging costs, transformer overload is caused by the charging power of all users in the charging stations, including the collective charging power of EVs and non-EVs load. Similar to Silva et al. (2020a) and Yaodong Yang and Yu (2019), this paper assumes that all charging station users interact directly with the distribution network via a transformer. Each agent can obtain the real-time load information of the transformer. The penalty for transformer overload can be defined based on the contribution of each household to the total power demand.  $\beta$  is the penalty factor for transformer overload, which is a big positive number, and a larger value indicates severer transformer overload penalty. Its value is determined empirically in practice.  $P_{overload,i}^t$  is the power that exceeds the transformer capacity caused by EV  $i$  at time step  $t$ , which can be expressed by Eq. (7):

$$P_{overload,i}^t = (a_i^t - p_{pv,i}^t) + \sum_{\substack{j=1 \\ j \neq i}}^M (a_j^t - p_{pv,j}^t) + P_{base,t} - C_{batCap} \quad (7)$$

where  $a_i^t - p_{pv,i}^t$  is the charging power of the current EV from the grid,  $\sum_{j \neq i}^M (a_j^t - p_{pv,j}^t)$  is the charging power from the grid for all current EVs except oneself,  $P_{base,t}$  is the base load, and  $C_{batCap}$  is transformer safety capacity. The term  $\tau \delta |e_{max} - e_t|$  is a measure of users' "range anxiety", the fear that at the end of their charging time an EV's energy will not be sufficient to reach their destination, a term proposed to penalize the number of EVs that fail to meet users' battery needs within a specified period of time. In order to measure the user of preference for the goal of saving cost and reducing distance anxiety, the anxiety coefficient  $\tau$  is introduced, so that the range anxiety term has the same unit of measurement as the charging cost term.  $e_{max}$  is the desired maximum SoC of the battery.  $e_t$  is the SoC when EV leaves.  $\delta_i^t$  is a decision variable, which is defined as Eq. (8):

$$\delta_i^t = \begin{cases} 0 & \text{if } t \neq t_{dep,i} \\ 1 & \text{otherwise} \end{cases} \quad (8)$$

where  $t_{dep,i}$  is the departure time of EV  $i$ .

### 3. Proposed approach

In this section, we first describe the mechanism of DDQN-PER and the tricks used to improve its performance; then discuss how to solve the EV charging/discharging problem formulated in Section 2 using DDQN-PER; finally, the EV charging/discharging algorithm based on DDQN-PER is proposed.

#### 3.1. DDQN-PER

In Section 2, the EV charging/discharging scheduling problem is formulated as a MDP problem, which can be solved in various ways, e.g., dynamic programming, Monte Carlo method, and temporal difference method. Q-learning (Sutton and Barto, 2018), a typical reinforcement learning method of time difference, relies on a Q table to store Q values. The agent selects the action that corresponds to the maximum

Q value in Q Table. For state  $s_t \in S$  and action  $a_t \in A$ , the Q-value of this state-action is determined by

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_t + \gamma \max_a Q(s_{t+1}, a_t) - Q(s_t, a_t)) \quad (9)$$

Where  $\alpha$  is the learning rate.  $\gamma$  is the discount factor and  $r_t$  is the reward obtained by taking action  $a_t$ .

However, when the dimensions of state space  $S$  and action space  $A$  are huge, the traditional Q learning method becomes infeasible due to the dimension disaster. To solve this problem, DRL introduces deep neural network (DNN) (Mnih et al., 2014) to approximate the action value function. The DQN outputs a vector of the action value  $Q(s_t, \cdot; \theta)$  based on the state  $s_t$ , where  $\theta$  is the trainable parameter of the DNN. Two important concepts are used in the DQN, namely, the target network and the experience replay. The parameters of target network are updated using the parameters of policy network every B steps and remain unchanged before next update. For experience replay, there is a dedicated replay memory to store the transformations observed during learning. The stored transformations are sampled to update the target network. The q-values in DQN are determined by

$$\begin{cases} Q(s_t, a_t, \theta) \leftarrow Q(s_t, a_t, \theta) + \alpha \tilde{x}_t - Q(s_t, a_t, \theta) \\ \tilde{x}_t = r_t + \gamma \max_a Q(s_{t+1}, a_{t+1}, \theta^-) \end{cases} \quad (10)$$

The maximization operator in DQN leads to the TD target overestimation. The target network can alleviate the bias caused by bootstrapping, but does not help to alleviate the overestimation caused by maximization. In order to separate action selection from evaluation during neural network training, a DDQN deep reinforcement learning method is proposed in Greff et al. (2017). In DDQN, policy networks select actions based on greedy algorithms, and the improved target network is used to estimate the value of the actions, which can alleviate the overestimation caused by maximization. The Q-value of this state-action function in DDQN is updated as

$$\begin{cases} Q^{new}(s_t, a_t, \theta) \leftarrow (1 - \alpha) \cdot Q(s_t, a_t, \theta) + \alpha \tilde{y}_t \\ \tilde{y}_t = r_t + \gamma Q(s_{t+1}, \arg \max_a Q(s_{t+1}, a; \theta); \theta^-) \end{cases} \quad (11)$$

To stabilize the training process of DNN, the experiences stored in replay memory are sampled using a uniform random sampling method. However, the importance of these experiences for training is not equal. To this end, the prioritized experience replay mechanism is proposed by Schaul et al. (2015), which samples experiences from replay memory according to their time difference (TD) errors, the experiences with larger TD error are more likely to be sampled. Compared to original experience replay method, it is more effective and can accelerate the training process of DNN. For each experience  $z_j \in Z$ , where  $Z$  denotes the set of stored experiences in replay memory. The TD error is expressed as follows:

$$\delta_{z_j} = R_{z_j} + \gamma_{z_j} Q_{target}(S_{z_j}, \arg \max_a Q(S_{z_j}, a)) - Q(S_{z_j-1}, A_{z_j-1}) \quad (12)$$

Where  $R_{z_j}$  is the reward for experience  $z_j$ ,  $\gamma_{z_j}$  is the discount factor. The priority of experience  $z_j$  is computed by

$$\chi_{z_j} = |\delta_{z_j}| + \varsigma \quad (13)$$

Where  $\varsigma$  is a small positive real number, which is used to guarantee every experience has a chance to be sampled even if its TD-error is zero. The capacity of experience replay memory is denoted by  $K$ . At time step  $t$ , the maximal priority of all experiences stored in replay memory is determined by Eq. (14):

$$\chi_{max,t} = \max_{i < t} \chi_{z_i} \quad (14)$$

where  $\chi_{z_j} > 0$  is the priority of experience  $i$ .  $\chi_{z_j}$  is the biggest priority at the current moment  $t$ . The probability of the experience  $z_j$  being sampled is determined by Eq. (15):

$$P(z_j) = \chi_{z_j}^\alpha / \sum_i \chi_{z_i}^\alpha \quad (15)$$



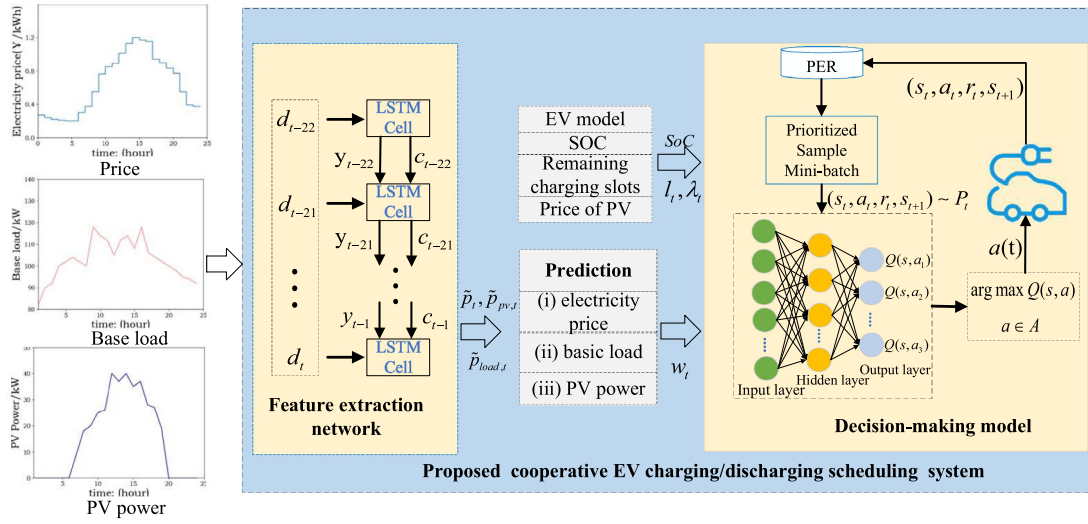


Fig. 2. The proposed algorithm process of EVs cooperative charging/discharging scheduling framework.

where  $P(z_j)$  is the sampling probability for experience  $z_j$ ,  $\sum_i \chi_{z_j}^\alpha$  is the total priority values, the exponent  $\alpha$  determines how much prioritization is used, with  $\alpha=0$  corresponding to the uniform case. A new data structure called SumTree is used to implement the experience replay memory (Schaul et al., 2015). It is a typical binary tree where the value of each node is equal to the sum of the node correlation values in its left and right subtrees. Then the importance-sampling weight  $w_{z_j}$  is obtained by Eq. (16):

$$w_{z_j} = (K \cdot P(z_j))^{-\zeta} \quad (16)$$

$$w'_{z_j} = w_{z_j} / \max_i w_{z_i} \quad (17)$$

that fully compensates for the non-uniform probabilities  $P(z_j)$  if  $\zeta=1$ . For stability reasons, the normalized weights  $w'_{z_j}$  is normalized by (17) so that they only scale the update downwards.

### 3.2. EV charging/discharging scheduling algorithm based on DDQN-PER

The overall framework of the proposed optimal EV charging/discharging strategy based on DDQN-PER is shown in Fig. 2. It mainly includes the feature extraction network and the decision model. Due to the high uncertainty of electricity price, PV power and real-time load, the feature extraction network can extract the discriminative features from the changing trends of these uncertain factors. Then, the features are concatenated with the SOC, the remaining charging time slot and the electricity price of PV. Finally, the features are sent to the Q network for learning to obtain the action values of all schedules in the current input state. The action with the largest Q value is then selected for execution.

#### A. Architecture of Feature extraction layer

(1) Feature extraction layer: Extracting feature information from raw time series data is a key step in action value function approximation. The Features should include the change trend of electricity prices, PV power, and transformer base load. Using these characteristics, scheduling strategy can minimize charging costs, utilize renewable energy as much as possible, and avoid transformer overload.

Recently, since LSTM-based neural networks (Greff et al., 2017) are relatively easy to implement and have good prediction performance, they have been widely used in predicting electricity price and load (Jahangir et al., 2021; Kong et al., 2019). Because the fluctuation of electricity price, PV output and base load have a quasi-periodic and natural time series, the change trend characteristics information can be obtained through the LSTM network. In the feature extraction network of Fig. 1, the trend of electricity price, PV output and transformer base

#### Algorithm 1 Predictive training process based on LSTM

**Input:** Electricity price  $p_{t-24}, \dots, p_{t-1}$ , PV output  $p_{pv}^{t-24}, \dots, p_{pv}^{t-1}$ , base load  $p_{load}^{t-24}, \dots, p_{load}^{t-1}$

**Output:** The forecasted electricity price  $\bar{p}_t$ , PV output  $\bar{p}_{pv}$  and basic load  $\bar{p}_{load}$  of current time.

1: **for** episode = 1:N **do**

2:  $\diamond \bar{p}_t = \text{LSTM}_1(p_{t-24}, \dots, p_{t-1})$   
 $\bar{p}_{pv} = \text{LSTM}_2(p_{pv}^{t-24}, \dots, p_{pv}^{t-1})$   
 $\bar{p}_{load} = \text{LSTM}_3(p_{load}^{t-24}, \dots, p_{load}^{t-1})$

3:  $\diamond Z^{\bar{p}_t} = (\bar{p}_t - p_t)^2$   
 $Z^{\bar{p}_{pv}} = (\bar{p}_{pv} - p_{pv})^2$   
 $Z^{\bar{p}_{load}} = (\bar{p}_{load} - p_{load})^2$

4:  $\diamond$  Updating the parameters of  $\text{LSTM}_1$ ,  $\text{LSTM}_2$  and  $\text{LSTM}_3$  to minimize the  $Z^{\bar{p}_t}$ ,  $Z^{\bar{p}_{pv}}$  and  $Z^{\bar{p}_{load}}$

5: **end for**

load are obtained through multiple LSTM networks. Specifically, the input is the past 24 h of electricity price, PV output and basic load information, and the output is the current one hour (t) electricity price, PV power and the transformer base load prediction of the charging station respectively. The predicted information is then fed into the DDQN-PER-based method to coordinate the charging/discharging of different EVs.

Algorithm 1 summarizes the prediction training process based on LSTM. The input of LSTM network is the electricity price  $p_{t-24}, \dots, p_{t-1}$ , PV power  $p_{pv}^{t-24}, \dots, p_{pv}^{t-1}$ , and transformer base load  $p_{load}^{t-24}, \dots, p_{load}^{t-1}$ . The output of LSTM network is the forecasted information of the electricity price, PV power and transformer base load. The LSTM networks are trained in a traditional supervised manner. Obtain the predicted price, PV power and transformer base load at step 2, calculate the loss with the label data at step 3, and update the parameters of the three LSTM neural networks at step 4.

#### B. Training of EVs Cooperative Charging Scheduling Algorithm

The EV charging scheduling algorithm is shown in Algorithm 2, the input is electricity price, SOC, remaining charging time slot, PV power, transformer base load. The output is the optimal charging/discharging power for EV. Firstly, the DNN's parameters  $\theta$  are randomly initialized. The target parameters  $\theta^-$  are copied from  $\theta$  in line 2. Each epoch starts at  $t_0$  and ends at the remaining charging time slot  $l_t=0$ . The initial state  $s_{t_0}$  is first obtained. Then on line 5, the EV charge and discharge action is performed from  $t_0$  to  $t_0 + l_t$ . In each time slot, the charging action  $a_t$  is chosen according to the greedy strategy,

**Algorithm 2** Training of cooperative EV charging/ discharging scheduling

**Input:** Prices, battery SOC, remaining slots, PV power, base load, capacity of replay memory  $K$ , total episodes  $T$  and reward  $r$   
**Output:** Schedule of charging/discharging time sequence.  
1: Initialize DNN's parameters  $\theta$   
2: Initialize target network parameters =  $\theta^- = \theta$ .  
3: **for** Epoch=1: $T$  **do**  
4:   Get the initial environment  $s_t$ .  
5:   **for** Time step  $t = t_0 : t_0 + l_t$  **do**  
6:     Choose action  $a_t$  according to  $\epsilon$ -greedy.  
7:     Execute action  $a_t$ , from (4) observe reward  $r_t$ , and from current state to the next state  $s_{t+1}$ .  
8:     Store experiences  $(s_t, a_t, r_t, s_{t+1})$  in  $K$  with maximal priority (14).  
9:     **if**  $t \equiv 0 \bmod K$  **then**  
10:       **for**  $j = 1$  to  $k$  **do**  
11:         Compute TD-error (12).  
12:         Update transition priority (13).  
13:         Sample minibatch of transitions (15).  
14:         Compute importance-sampling weight (16) and (17).  
15:         Accumulate weight-change  $\Delta \leftarrow \Delta + w_j \cdot \delta_j \nabla_{\theta} Q(S_{j-1}, A_{j-1})$   
16:       **end for**  
17:       Update weights  $\theta \leftarrow \theta + \eta \cdot \Delta$ , reset  $\Delta = 0$ .  
18:       From  $B$  steps copy weights into target network  $\theta^- \leftarrow \theta$ .  
19:     **end if**  
20:     Choose action  $A_t \sim \pi_{\theta}(S_t)$ .  
21:   **end for**  
22: **end for**

then execute the action and obtain reward  $r_t$ , then change to a new environment state. In line 8, each scheduling experience  $(s_t, a_t, r_t, s_{t+1})$  is stored in the replay memory with the maximum priority  $\chi_{\max,t}$ ; if the replay memory is full, the early samples will be removed, and the latest training samples will be stored. Then, TD-error is calculated in line 12, and the priority of experiences is updated in line 13. A small batch of charging/discharging experiences were extracted from the experiences replay memory, the weight of importance sampling in line 14 and 15, and accumulated weight-change in line 16. The small-batch gradient descent method is used to update the parameters  $\theta$  of DNN network. At every  $B$  epochs, the target network parameters are updated as  $\theta^- = \theta$ . At the end of the training epochs, the parameters  $\theta$  will be output for real-time online charging/discharging scheduling of EV.

Next, The Q network calculates action-value  $Q^{new}(s_t, a_t, \theta)$  from (11). And the EV charging/discharging action  $a_t$  is selected as  $a_t = \arg\max_{a \in A} Q^{new}(s_t, a_t, \theta)$ . The training process ends when the remaining charging time slot is 0. The optimal strategy for each step is represented by a probability distribution function  $\pi_{\theta}(s_t | \theta_{\pi})$ , and in each step according to the probability distribution is sampled to obtain the current best action value; that is:  $a_t \sim \pi_{\theta}(s_t | \theta_{\pi})$ . The process of generating a action is essentially a stochastic process. Finally, the charging/discharging scheduling timing arrangement  $a_{t_0:t_0+l_t}$  is outputted.

## 4. Experiment results

In this section, the detailed experimental parameter settings are given firstly, then evaluate the performance of EV charging/discharging scheduling algorithm based on DDQN-PER in multiple cases.

### 4.1. Experiment setup

This paper takes a charging station in an industrial park with EV access and PV power generation system as an example for simulation calculation in five typical working days. It is assumed that the industrial park is supplied by a transformer with a rated capacity of 160 kVA. Generally, the ideal limit (Anon, 2019) is when the transformer load

**Table 1**

EV user behavior parameters.

| Parameters      | Distribution             | Boundary                |
|-----------------|--------------------------|-------------------------|
| Arrival time    | $t_A \sim N(8, 1^2)$     | $6 \leq t_A \leq 11$    |
| Remaining slots | $l_t \sim N(8, 1^2)$     | $6 \leq l_t \leq 11$    |
| Battery SOC     | $SOC \sim N(0.5, 0.1^2)$ | $0.2 \leq SOC \leq 0.9$ |

**Table 2**

Simulation parameters.

| Parameters                                  | Value   |
|---------------------------------------------|---------|
| Simulation of days                          | 5       |
| Transformer capacity $C_{batCap}$           | 160 kVA |
| Price of PV $\lambda_t$                     | 0.45    |
| Cost of battery $C_{batCost}$               | 25000   |
| The slope of the linear approximation $m_k$ | 0.015   |
| Number of training steps $T$                | 80000   |
| $\epsilon$ -greedy $\epsilon$               | 0.95    |
| Discounted rate $\gamma$                    | 0.95    |
| Learning rate $\alpha$                      | 0.0001  |
| Mini-batch size $k$                         | 32      |
| Target network update frequency $B$         | 10000   |

ratio is 0.75, so the transformer safety limit is set as 120 kVA. In view of China's policies on renewable energy power generation, the electricity price of PV is set as 0.45RMB/kWh. The hourly electricity price data is obtained from the Nord-pool database in China. The comparison with ground truth and prediction values of electricity price, base load and PV power for 120 consecutive hours from Sept.1, 2019 at 00:00:00 to Sept.5, 2019 at 23:59:59 in the park is shown in Fig. 3(a), 3(b) and 3(c). By analyzing the commuting behavior of users in the industrial park, EV users usually arrive at around 08:00 and leave at around 18:00. EV users expect to drive away with a SOC higher than 80%, and a SOC not lower than 0.2 and not higher than 0.9 during charging/discharging process.

According to the suggestion of Yao et al. (2017b), because the travel of charging station users in the park has certain regularity, it can be modeled as random variables. Among them, the arrival time of the EV, the remaining charging time slot and the initial battery SOC at the charging station in the park obey the truncated normal distribution. The distribution is shown in Table 1. In this paper, EV battery capacity is 24 kWh, the charging/discharging power provided by the charging pile is set as discrete value, and set  $n$  as 3 which can be divided into seven levels ( $-6$  kW,  $-4$  kW,  $-2$  kW,  $0$  kW,  $2$  kW,  $4$  kW,  $6$  kW). The time slot  $\Delta t$  is 1 h, and it is assumed that the charging/discharging power remains unchanged within the time interval.

### 4.2. Model training process

The detailed experimental settings of training process are set as shown in Table 2. The learning rate  $\alpha$  in the training process is set as 0.0001, and the discount rate  $\gamma$  and  $\epsilon$  are set as 0.95,  $m_k$  is set to 0.015. China clearly proposes that the production price of batteries should be reduced to 1 RMB/W.h in 2020 (Zhang et al., 2022), thus setting the battery replacement cost to 25000 RMB, and the penalty factors  $\tau, \beta$  is set as 0.2, which is conducive to the convergence of training by controlling all rewards and punishments within the same range.

### 4.3. Results analysis

The training epoch starts when the EV enters the charging pile and ends when the remaining charging time of  $l_t$  is 0. In each epoch, the cumulative reward values can be calculated by the  $\sum_{t=t_0}^{t_0+l_t} r_t$ . The evolution of the cumulative rewards over 80,000 epochs is illustrated in Fig. 4, the abscissa is the number of training rounds, without the unit axis, the ordinate is the cumulative reward value, the unit is RMB. In the first 20,000 sets of training, the agent charge and discharge

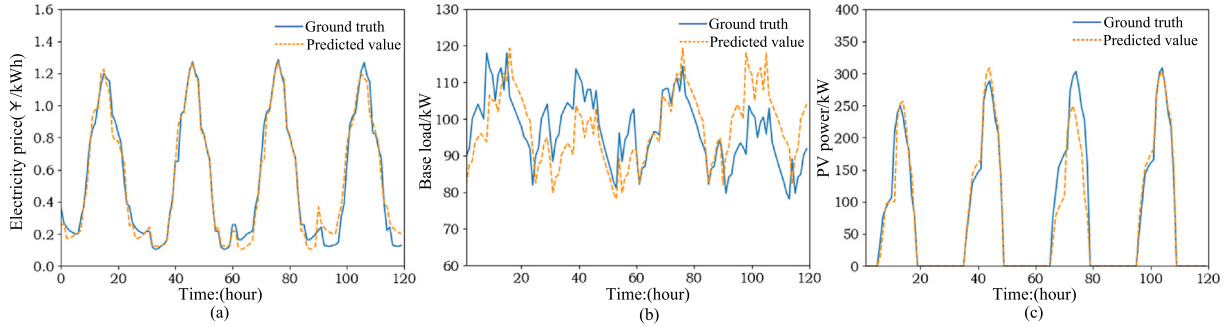


Fig. 3. Predicted results: (a) Electricity price, (b) Base load and (c) PV power forecasting for days from Sept.1, 2019 at 00:00:00 to Sept.5, 2019 at 23:59:59.

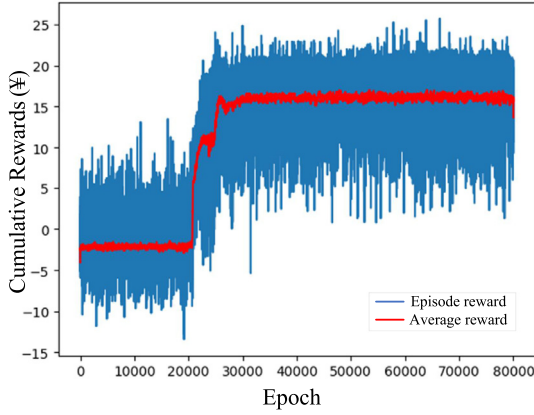


Fig. 4. The changes in cumulative rewards over the course of training.

behavior is completely randomly selected, so that the agent can fully explore the environment, which is why the start time reward varies greatly. Then, after completing a certain degree of training, the agent exploration rate  $\alpha$  is reduced to 0.0001, and the agent begins to use the knowledge accumulated so far to self-learn to the optimal charge-discharge action. We can observe that around 37,000 rounds, cumulative rewards begin to converge at around 15. This means that EVs have learned to map the state space of a given input to an optimal charging scheduling arrangement for actions with a maximum cumulative reward.

In order to investigate the performance of the cooperative charging/discharging scheduling algorithm, the charging/discharging power of a EV for 5 consecutive days is shown in Fig. 5. The electricity price (RMB/kWh) is represented by blue line, and the charging/discharging energy (kWh) is represented by the red bar. As shown in Fig. 5, agents can perform charging actions when the price is low, and perform discharge actions when the price is high. The corresponding battery SOC of these five consecutive days is shown in Fig. 6. During the five days of testing, the battery SOC reached 0.83 at the end of charging, which can meet the charging demand of users. Through increasing the coefficient  $\beta$  of the reward function, a higher average final SoC can be obtained, which in turn increases the penalty for uncharged EVs.

In order to show the relationship between EV charging/charging power and PV energy output, one day is randomly selected from these five days. The predicted electricity price, transformer base load, and PV energy output of the selected day are shown in Fig. 7. Fig. 8 shows the PV power output (blue line) and the charging/discharging power of EV (red bar). From Fig. 8, we can conclude that EV is more likely to charge when the PV output power is high, which is very helpful to improve the utilization rate of renewable energy and minimize the charging cost of EV user.

To further evaluate the performance of the proposed algorithm, it is compared with several benchmark solutions, including uncontrolled strategy, DDQN, DQN-PER, Q-learning. Uncontrolled strategy means that once an EV is plugged in, it will charge at its maximum charging power until it is fully charged. DDQN refers to the use of deep Q network algorithms to train agents to achieve maximum reward values. DQN-PER is to train agents through deep Q network with priority experience replay to obtain the optimal scheduling strategy. The Q-learning based algorithm is using Q table to train EVs to maximize the action value function. The average training times and charging costs for the proposed and baseline methods over the test days are listed in Tables 3 and 4, the negative value represents the profit from discharge of EV. The proposed DDQN-PER algorithm can not only greatly shorten the training time, but also achieve the minimal average cumulative charging cost compared with the DDQN, DQN-PER, Q-learning and uncontrolled algorithms. Different from other fields, in the EV charging/discharging scheduling field, there is no commonly used dataset, researchers usually validate their algorithms using different private datasets. It is difficult to directly compare the results with other published papers, to this end, the reduction rate of charging cost compared with uncontrolled charging strategy is selected as comparison indicator. The charging cost reduction rates of [12] and [18] are 77.3% and 75%, respectively. The average charging cost reduction rate of our proposed algorithm is 160%. Therefore, we can conclude the proposed algorithm has an obvious advantage in reducing EV charging cost.

To verify the robustness of the proposed algorithm, we selected 20 different sets of inputs (divided into four groups of A, B, C, and D) to conduct simulation experiments. As shown in Fig. 9, the average charging cost of the four groups is reduced by about 158%, the best value is 175%, the worst value is 135%, and the variance is 0.0155 compared with uncontrolled charging strategy. Fig. 10 shows the comparison of discharging profit between the proposed and several benchmark algorithm with or not PV, which proved that the user discharge profit can be effectively increased when PV is considered. The PV energy utilization rate under different algorithms is shown in Fig. 11, and we can see that the proposed algorithm has the best performance, and the average utilization rate can reach 82%. In summary, these results show that the proposed algorithm can minimize charging cost while satisfying user charging demand, facilitating PV energy utilization.

In order to further verify the advantage of proposed algorithm in avoiding transformer overload, five EVs are taken as objects to obtain transformer load values at 24 h of the day, and compared them with uncontrolled strategy, as shown in Fig. 12. That day, the EVs arrived at the charging station in the park at 9:00 am and left at 18:00 pm. During this period, if the uncontrolled strategy charging algorithm is used, that is, five EVs are connected and charged immediately. When the remaining transformer capacity is small, it will inevitably cause the peak load of the transformer reaches 130 kW, which is 8% higher than the safety limit of the transformer. However, the proposed algorithm has obvious advantages in reducing the peak load of the transformer compared with uncontrolled strategy. Because the EV charging sequence is optimized,

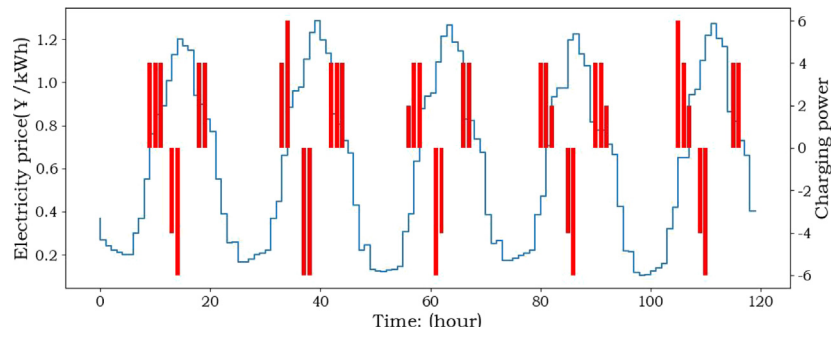


Fig. 5. Real-time electricity price curve (blue line) and charging/discharging power (red bar).

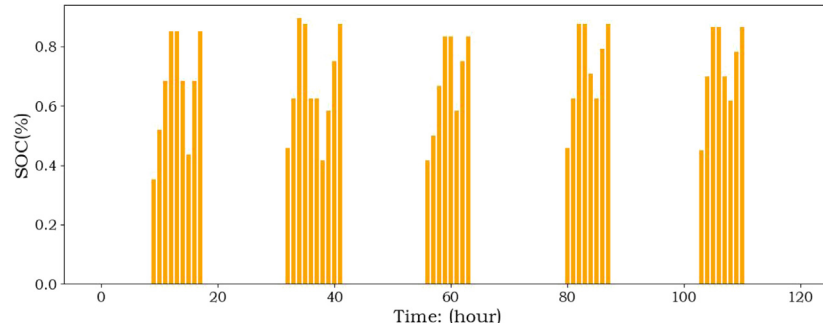


Fig. 6. SOC of EV over five consecutive days.

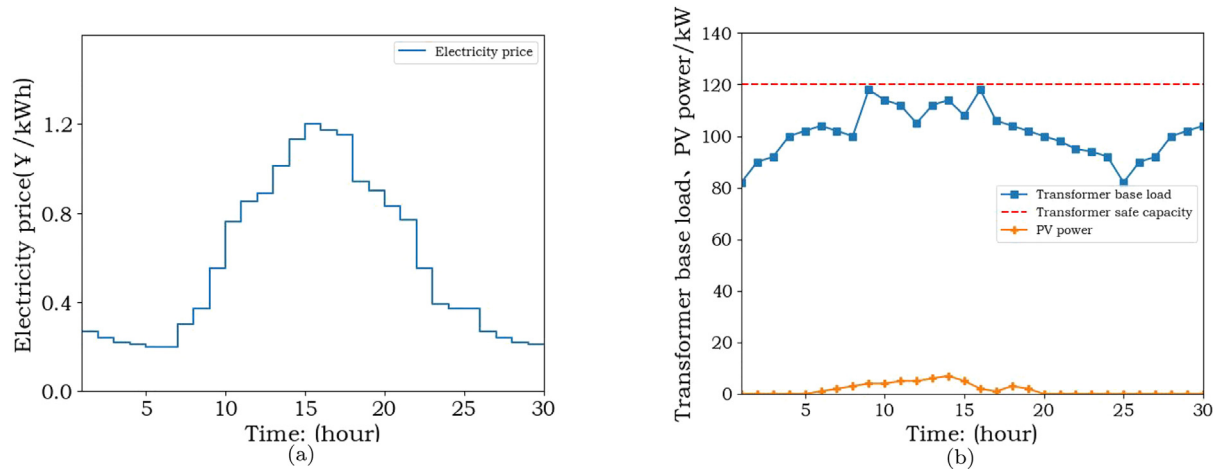


Fig. 7. Forecasted results of one day: (a)Electricity price, (b)PV power and Base load.

Table 3

Comparison of average training time of different scheduling algorithms.

| Algorithm              | DDQN-PER   | DDQN      | DQN-PER    | Q-learning |
|------------------------|------------|-----------|------------|------------|
| Training time(minutes) | 2 min 47 s | 3 min 4 s | 2 min 45 s | 30 min     |

Table 4

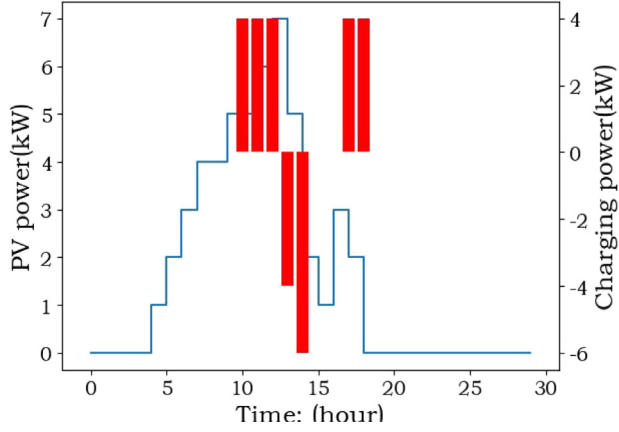
Comparison of charging cost of different scheduling methods.

| The simulation of day | DDQN-PER | DDQN  | DQN-PER | Q-learning | Uncontrolled |
|-----------------------|----------|-------|---------|------------|--------------|
| Charging cost/RMB     |          |       |         |            |              |
| 1                     | 0.67     | 0.04  | 2.76    | 0.83       | 4.20         |
| 2                     | -3.31    | -2.83 | -0.62   | 4.03       | 4.40         |
| 3                     | -3.11    | -2.97 | -0.77   | 4.23       | 5.80         |
| 4                     | -8.71    | -5.18 | -2.12   | 1.83       | 1.80         |
| 5                     | -9.90    | -7.34 | -6.15   | -5.0       | 3.00         |
| Average               | -4.88    | -3.67 | -1.38   | 0.98       | 3.84         |

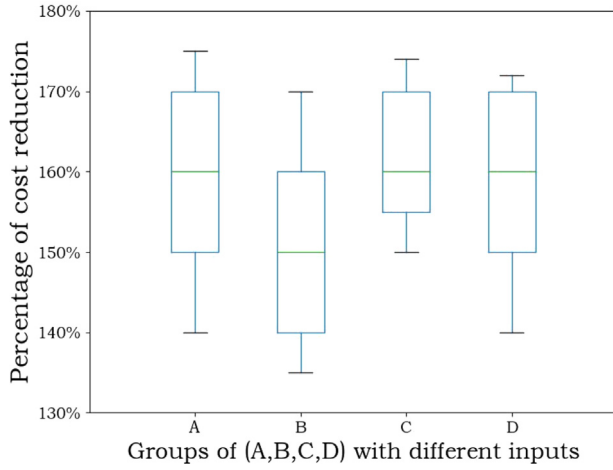


**Table 5**  
Three EVs user behavior parameters.

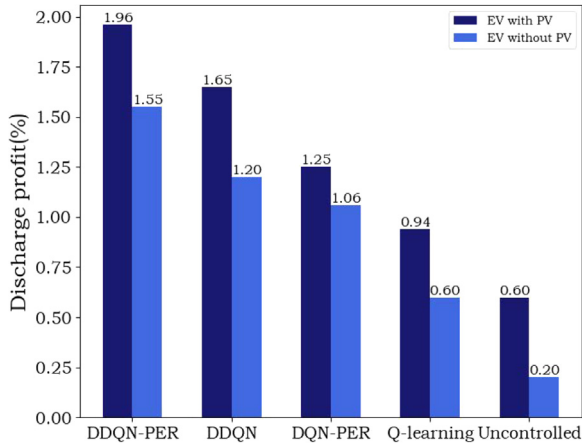
| Numbers | Arrival time | Departure time | SoC of arrival time | SoC of departure time | Battery capacity |
|---------|--------------|----------------|---------------------|-----------------------|------------------|
| EV1     | 8:00am       | 16:00 pm       | 0.30                | 0.9                   | 48 kWh           |
| EV2     | 9:00 am      | 17:00 pm       | 0.40                | 0.9                   | 48 kWh           |
| EV3     | 10:00 am     | 18:00 pm       | 0.45                | 0.9                   | 48 kWh           |



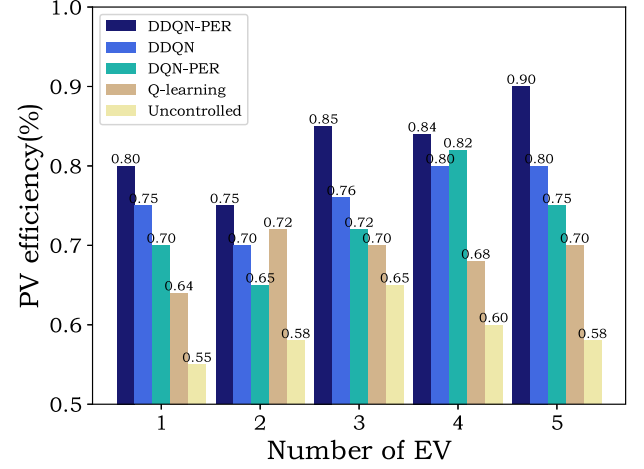
**Fig. 8.** Hourly PV power (blue line) and charging/discharging energy (red bar).



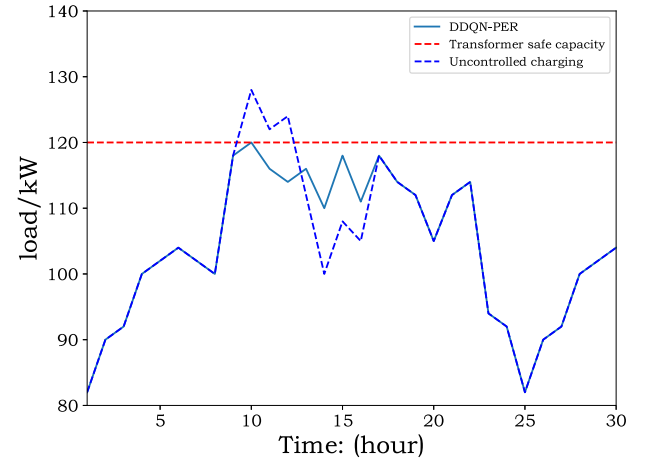
**Fig. 9.** The percentage of cost reduction compared with uncontrolled charging under 20 groups of different inputs.



**Fig. 10.** Comparison of discharge profits under different methods with or not PV.



**Fig. 11.** Comparison of PV efficiency under different methods.



**Fig. 12.** Comparison of Hourly transformer load between proposed algorithm and disordered charging.

the transformer total load in each time step is within its safe range. From the above experimental results, it can be seen that after the cooperative optimization scheduling, the utilization rate of PV power can be maximized and transformer overload can be avoided while the power demand of users can be met.

To prove the cooperation of the proposed charging network, three EVs are used to test the charging/discharging behaviors in uncontrolled mode and cooperative mode, respectively. The simulation parameters are listed in Table 5, and the safe capacity of transformer is set to be 120 kW. The simulation results are shown in Figs. 13–15. In uncontrolled mode, as shown in Fig. 13(a), EVs are charged at the maximum charging power, 6 kW, as soon as they are plugged in power grids, and the maximum total charging power is 18 kW. After charging, the EVs retain idle until EV users leave. As shown in Fig. 14, in uncontrolled mode, the total transformer load has exceeded the safe capacity at time slots 10 and 12. The total transformer load at time slot 10 is 124 kW, which exceeds the transformer safe capacity by about 3.3%. Different from the uncontrolled mode, in cooperative mode, EVs

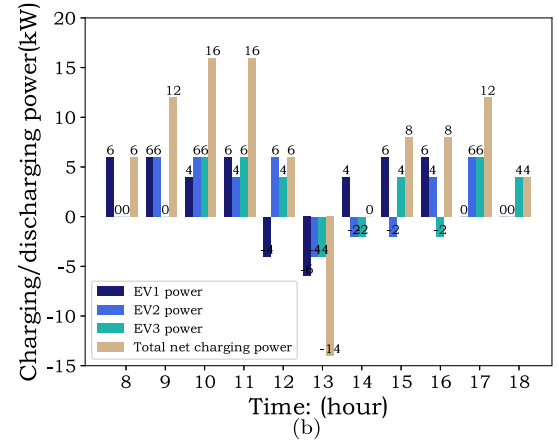
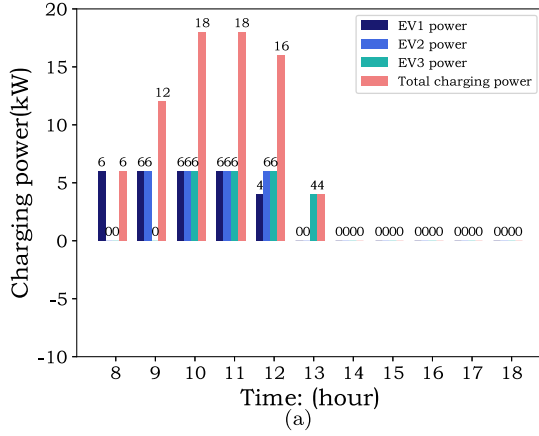


Fig. 13. Hourly charging power of three EVs and total power: (a)Uncontrolled strategy (b)The cooperative network.

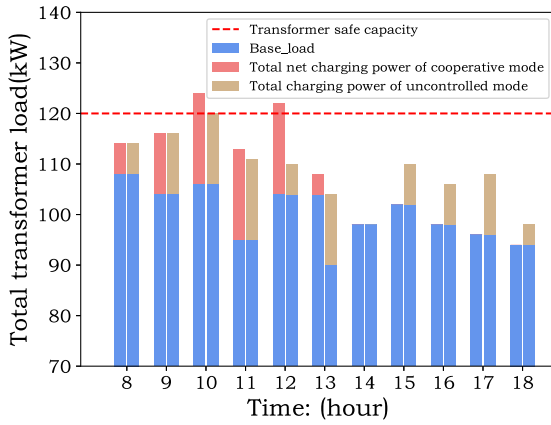


Fig. 14. Hourly transformer load of the cooperative network and uncontrolled strategy.

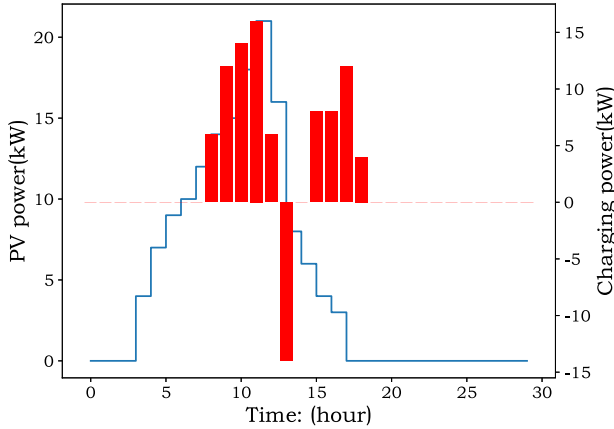


Fig. 15. Hourly PV power (blue line) and charging/discharging energy of three EVs (red bar).

can not only be charged by G2V(Grid-to-Vehicle) but also discharge the stored energy to grids by V2G(Vehicle-to-Grid). The maximum charging load is 16 kW, which is reduced by 11.11%, compared to the maximum charging load in uncontrolled mode. From Fig. 14, we can conclude the transformer is not overloaded throughout the whole scheduling process. In addition, at time slot 13, some base load is met by the energy discharged by EVs, and the net base load is reduced. At time slot 14, the charging power of EV1 is equal to the total discharging power of EV2 and EV3, and the net charging power is zero. The PV power output

and the total net charging power of EVs are shown in Fig. 15. From Fig. 15, it can find that EVs tend to charge when the PV power output is high, which helps to improve the utilization rate of renewable energy and reduce the charging cost. At time slot 14, although the PV power output is high, the electricity price is at its highest point, as shown in Fig. 7(a), when discharging EV is more profitable than charging.

## 5. Discussion

The efficiency of charging power is assumed to be 100%, which is the limitation of this study. The proposed charging scheduling network can be extended to learn a stable cooperative charging control policy for multiple EVs, and multiple renewable energy sources like wind turbines. This work currently studies the scheduling problem of a single charging station, and the future work will focus on extending the algorithm to the scheduling problem of multiple microgrids with multiple charging stations, then the demand response of charging station load, and selection of the charging/discharging power levels number will be also considered.

## 6. Conclusions

This paper studies the application of deep reinforcement learning to control the charging/discharging behavior of EVs. By considering the uncertainty of real-time electricity price, transformer base load and PV power generation, the improved DDQN method and the corresponding reward function are applied to solve the cooperative charging problem. On the premise of improving the utilization rate of PV power generation and reducing the peak value of the transformer, the charging demand of users is met, and the charging cost and battery degradation cost are the lowest. The performance of the proposed algorithm is evaluated by the simulations using real data. Experimental results show that this method can learn a stable cooperative charging control strategy. The advantages of the proposed algorithm in reducing charging cost, improving PV utilization rate, and avoiding transformer overload. The proposed charging scheduling network can be extended to learn a stable cooperative charging control policy for multiple EVs, and multiple renewable energy sources. Charging losses are neglected in this study, and extending the algorithm to the scheduling problem of multiple microgrids with multiple charging stations is the future work.

## CRediT authorship contribution statement

**Yanyu Zhang:** Conceptualization, Methodology, Writing, Reviewing, Writing – original draft. **Xinpeng Rao:** Conceptualization, Methodology, Writing, Reviewing, Writing – original draft. **Chunyang Liu:** Literature research, Writing – original draft. **Xibeng Zhang:** Supervision, Reviewing & editing. **Yi Zhou:** Supervision, Reviewing & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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